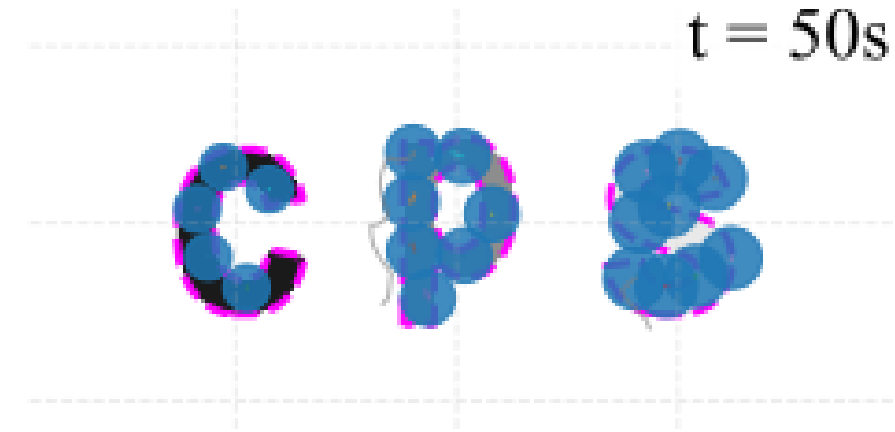
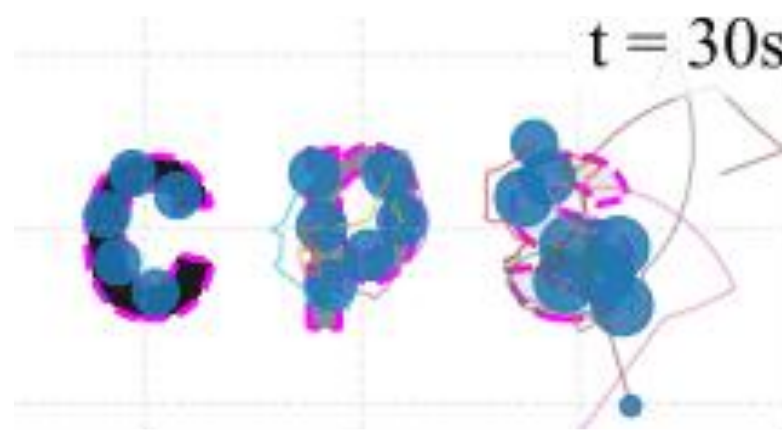
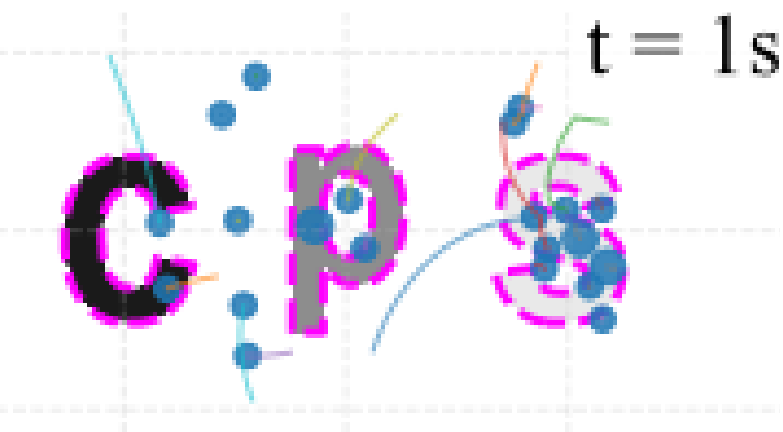
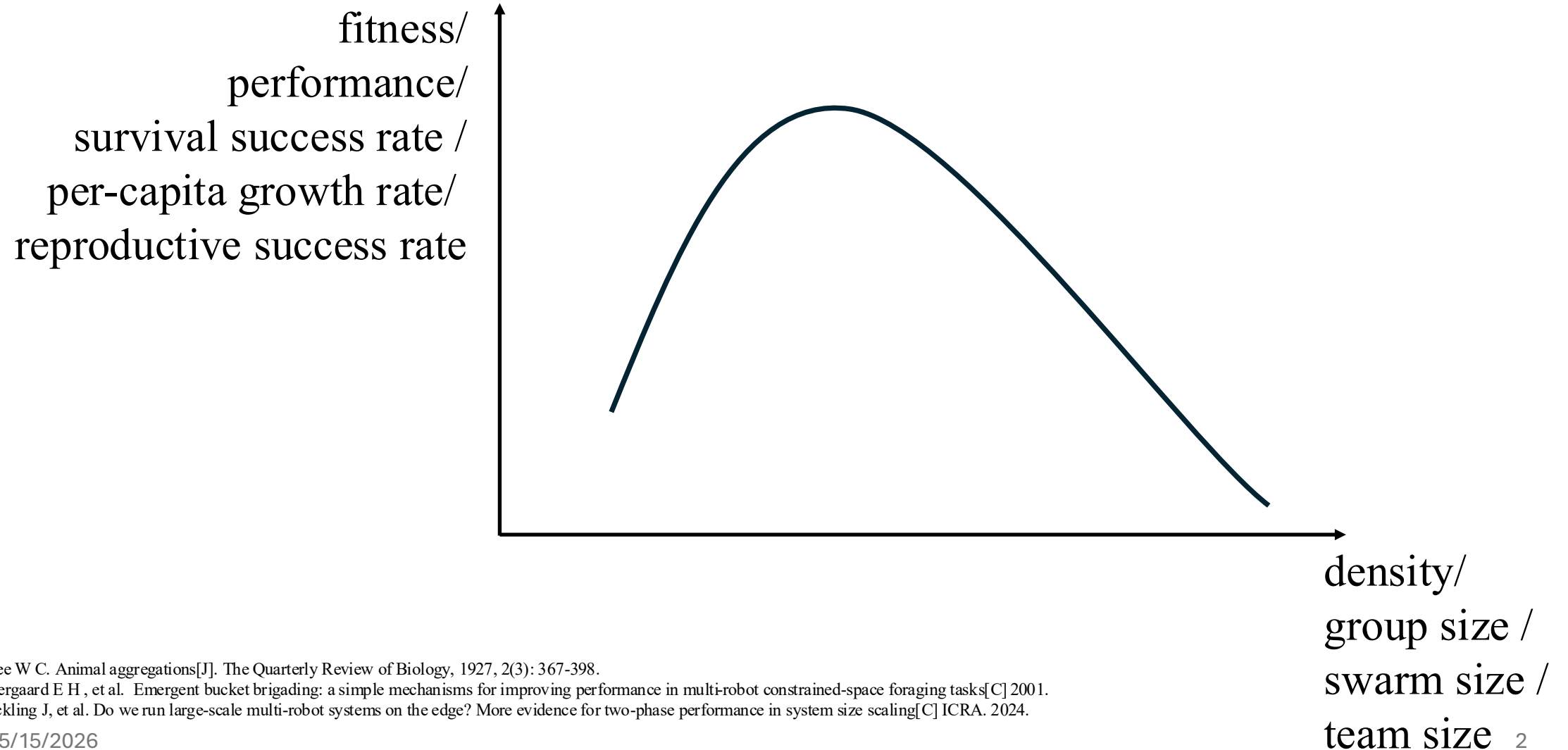
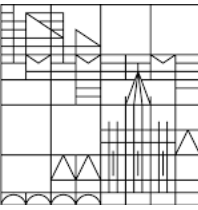


Fission–Fusion Processes for Regulation of Group Size and Number in Multi-Agent Systems

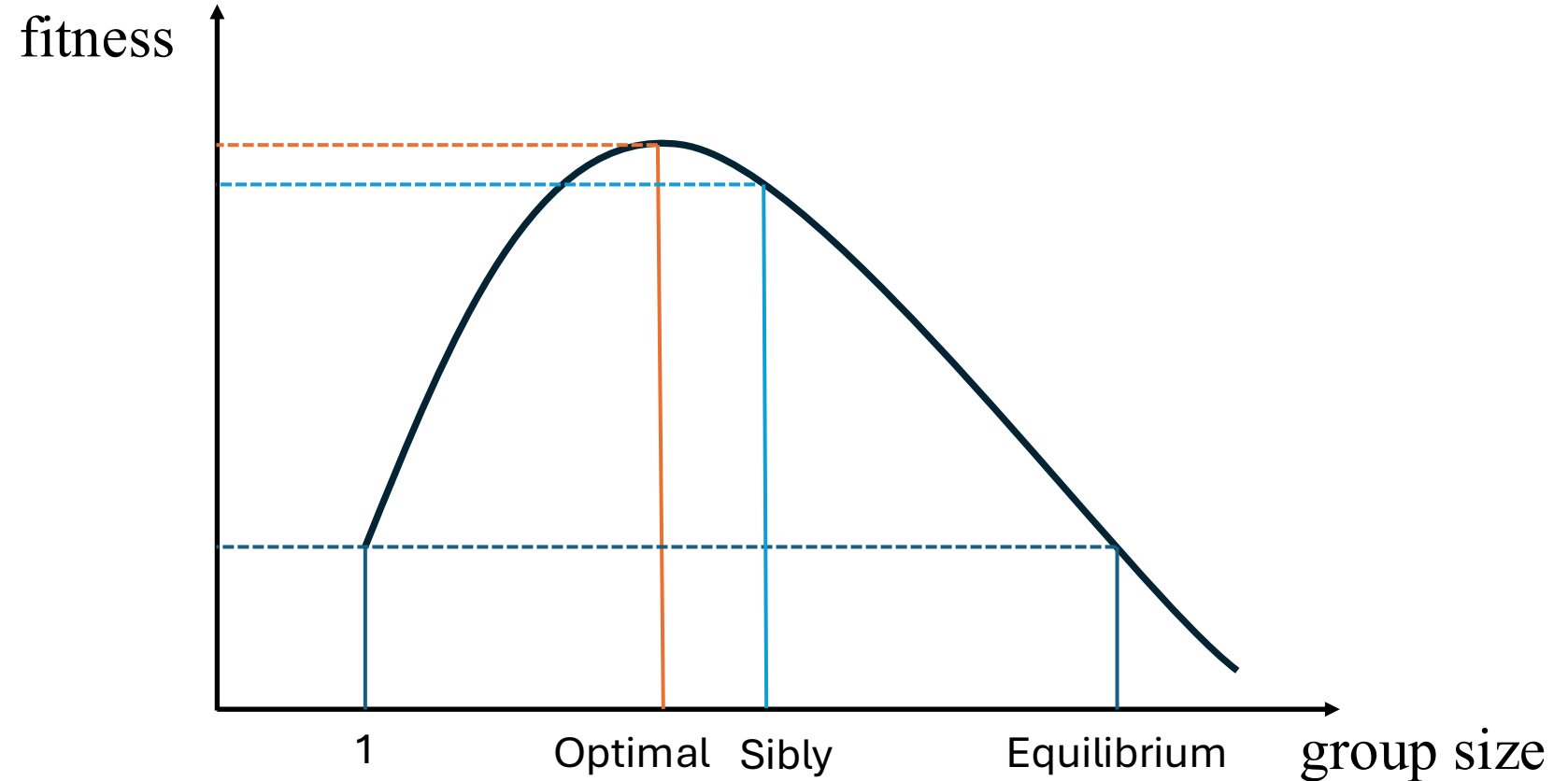
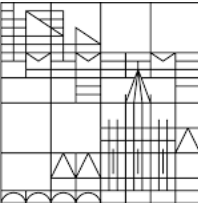
Tianfu Zhang
Cyber-Physical Systems Group



Inverted U-shaped curve



Optimal group size is unstable

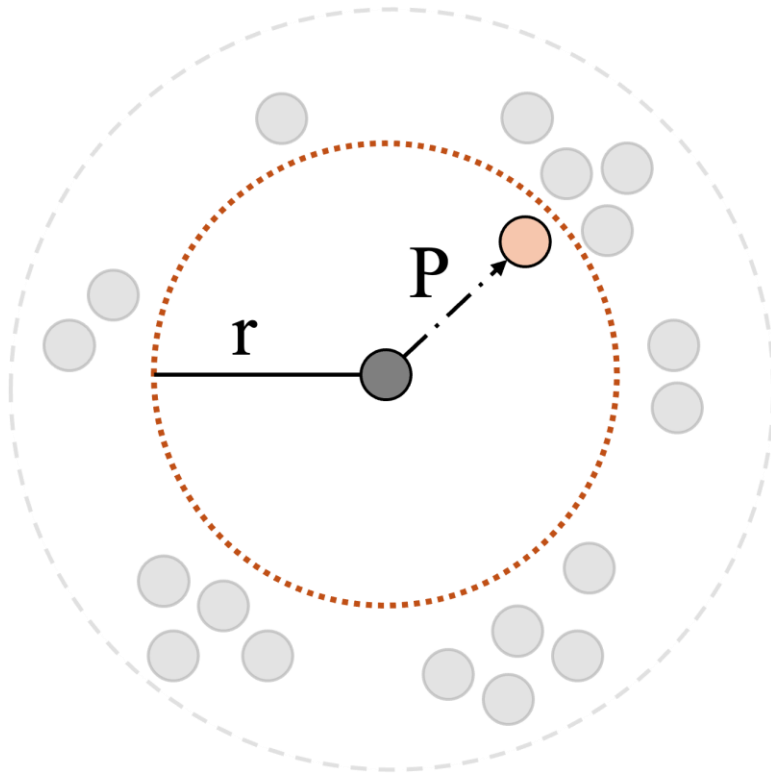
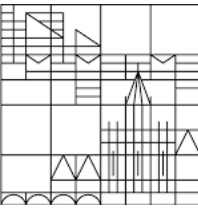


Sibly R M. Optimal group size is unstable[J]. Animal behaviour, 1983.

Beauchamp G , et al. The group-size paradox: effects of learning and patch departure rules[J]. Behavioral Ecology, 2005, 16(2): 352-357.

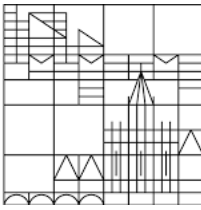
Aureli F, et al. Fission-fusion dynamics: new research frameworks[J]. Current anthropology, 2008, 49(4): 627-654.

Followerhip under constraints



- Follow probability P
- Follow range r

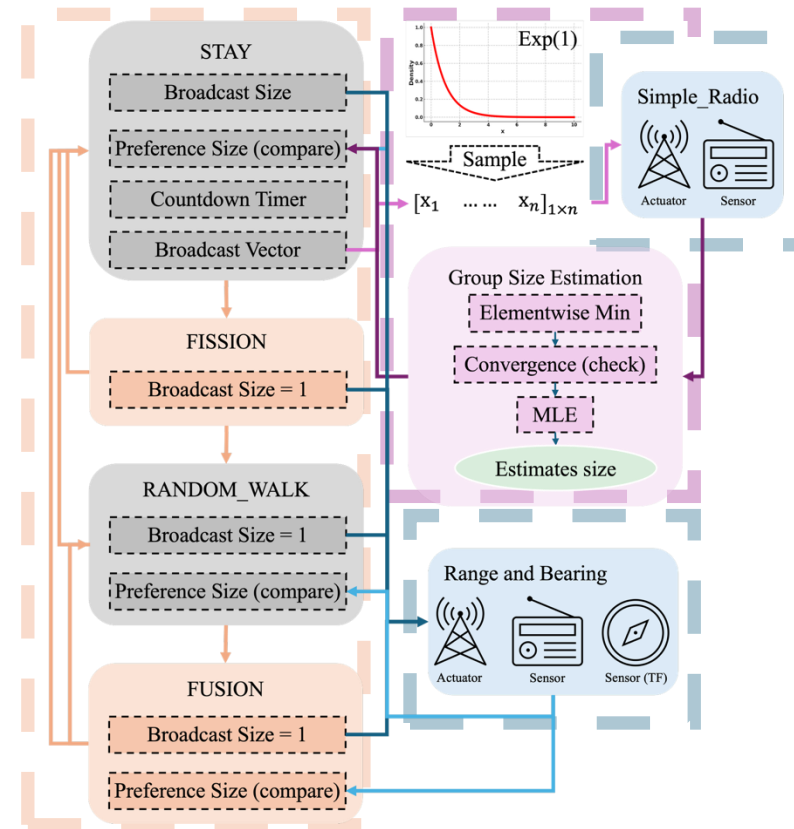
Overview



(3) Decision and Behavior Regulation Module

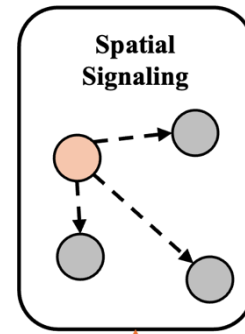
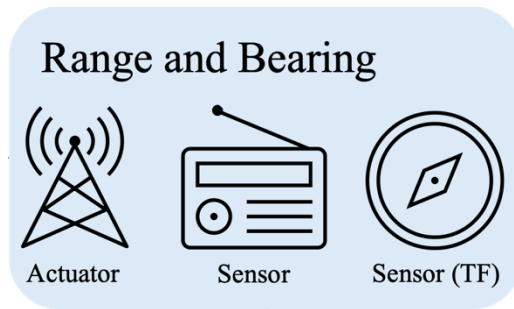
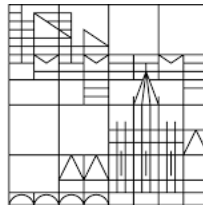
(2) Group Size Estimation Module

(1) Communication and Perception Module

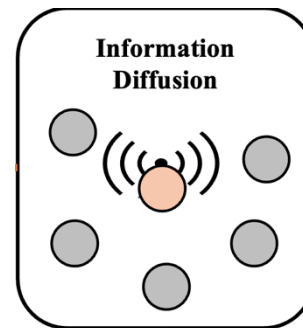
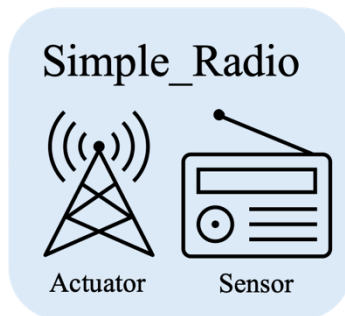


System Overview

(1) Communication

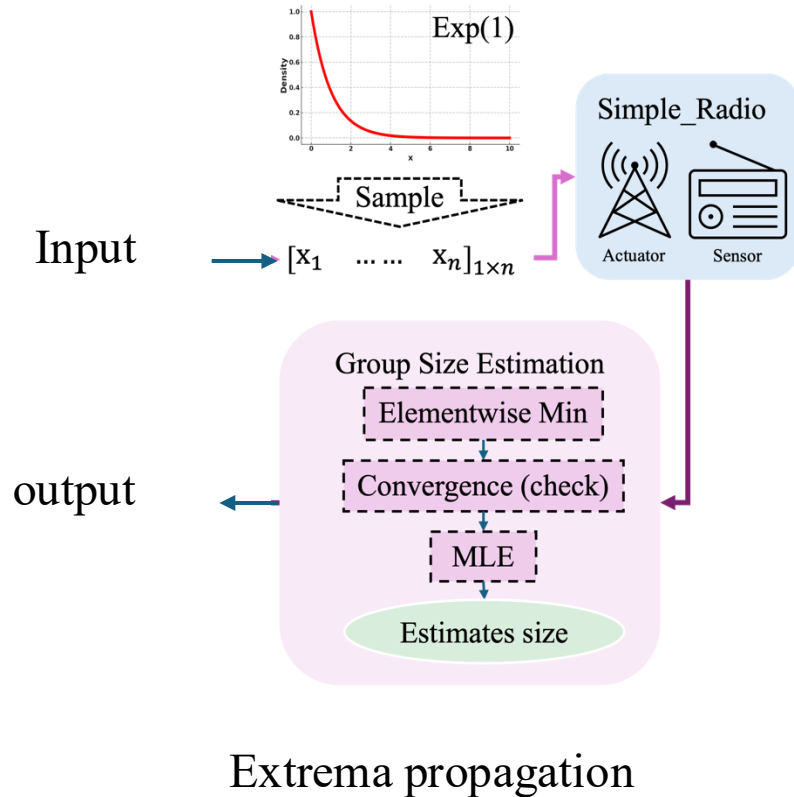
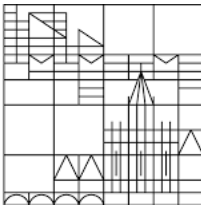


1. Positional Signaling:
2. Group-Size Signaling with one-shot decision
3. Group-Size Signaling with continuous decisions



- Round_id + numerical vectors

(2) Group size estimation



K- dimension:

$$X_i^k \sim \text{Exponential}(1), i = 1, \dots, N, k = 1, \dots, K$$

$$M_k = \min(X_1^k, X_2^k, \dots, X_N^k) \text{ each node: } (M_1, \dots, M_K)$$

$$M_k \sim \text{Exponential}(N)$$

1- dimension:

$$X_i \sim \text{Exponential}(1), i = 1, \dots, N$$

$$M = \min(X_1, X_2, \dots, X_N)$$

$$M \sim \text{Exponential}(N)$$

$$E[M] = \frac{1}{N}$$

$$\hat{N} = \frac{1}{M}$$

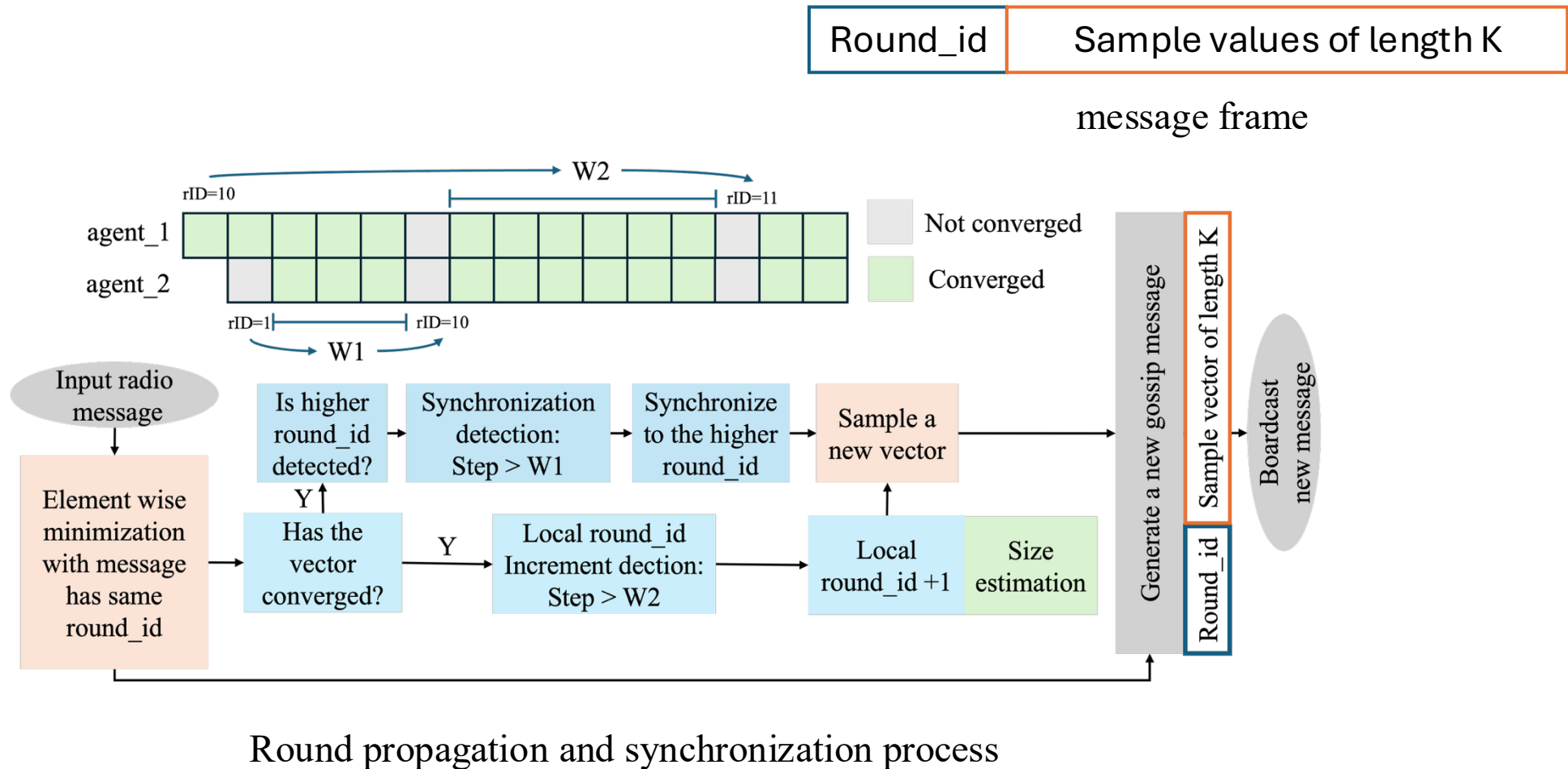
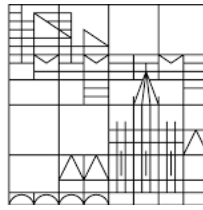
$$S_K = \sum_{k=1}^K M_k$$

$$S_K \sim \text{Gamma}(K, N)$$

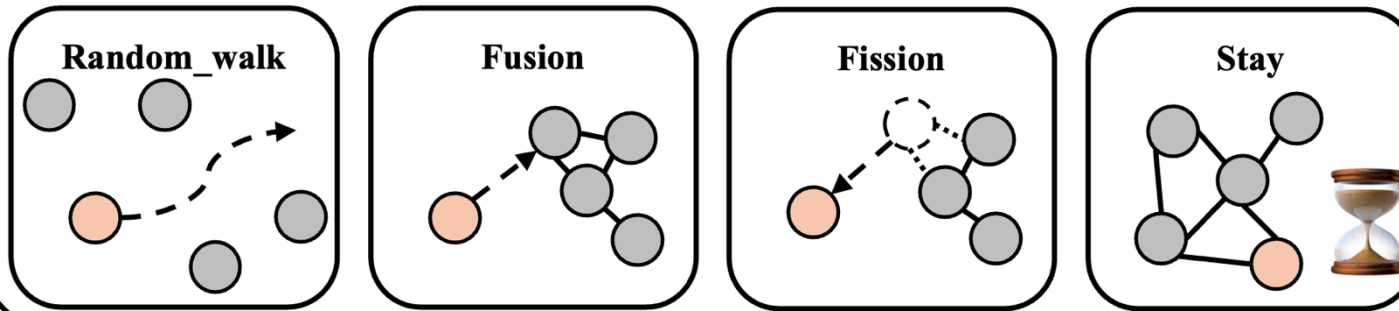
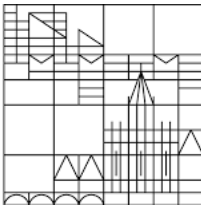
$$E[S_K] = \frac{K}{N}$$

$$\hat{N} = \frac{K}{\sum_{k=1}^K M_k}, \quad \text{Var}(\hat{N}) \propto \frac{1}{K}$$

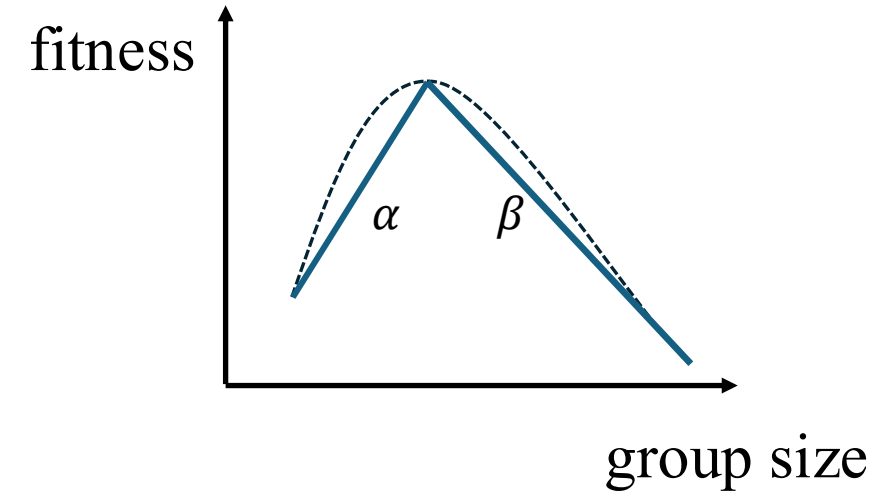
(2) Group size estimation



(3) Decision and behavior regulation



Current State	Random_Walk	Fusion	Stay	Fission
Random_Walk	–	find target	$P_{follow} < 1$	–
Fusion	$G^T > G^*$	–	close to target	–
Stay	waiting time exceeded	–	–	$G^E > G^*$
Fission	far from origin	–	–	–

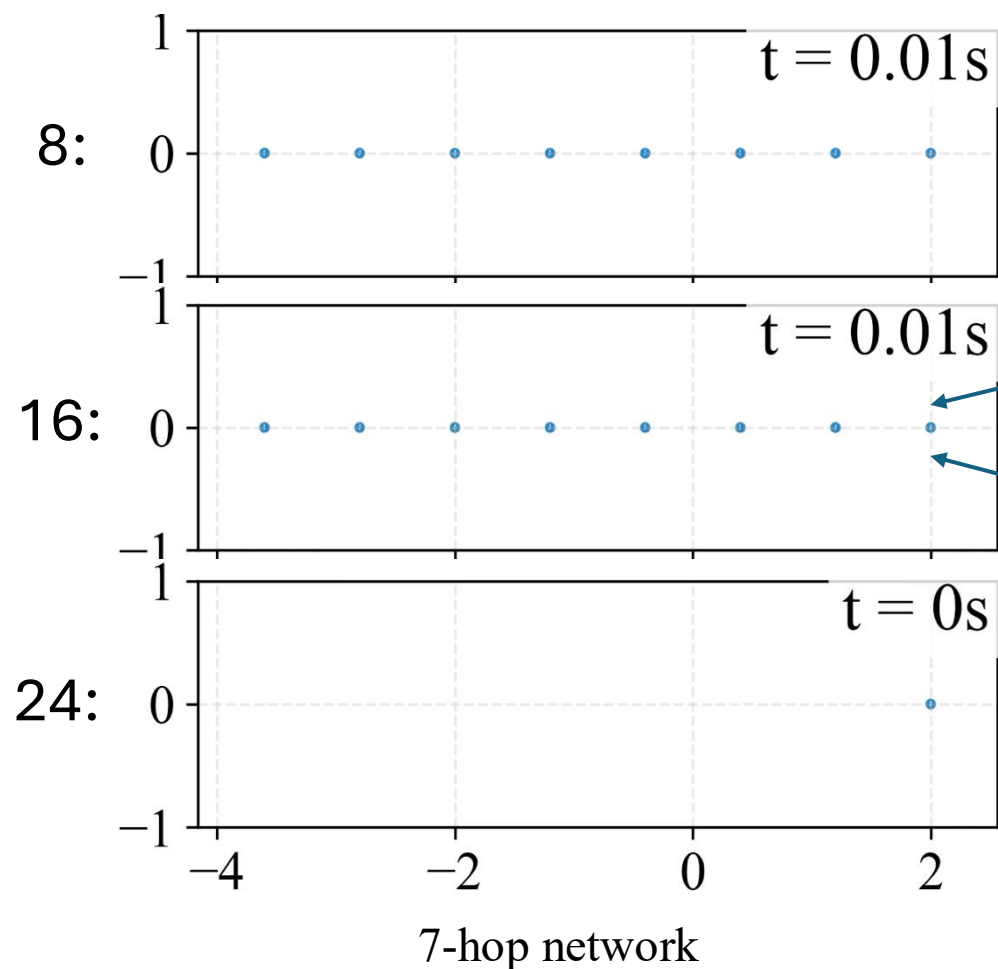
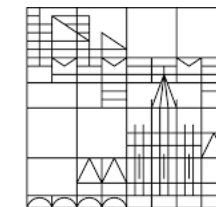


$$T = \alpha G_E + T_0$$

$$[P_{follow}, r] = \begin{cases} [P_{random_walk}, r_{random_walk}], & \text{if in random walk state} \\ [\beta * G^* / G_E, r_{stay}], & \text{if in stay state} \end{cases}$$

Gossip-based group-size estimation

- Fully decentralized: boardcasting
- Anonymous agents: doesn't need agent id
- Asynchronous operation: round synchronization
- No explicit point-to-point communication: radio
- Exact group-size estimation: not density

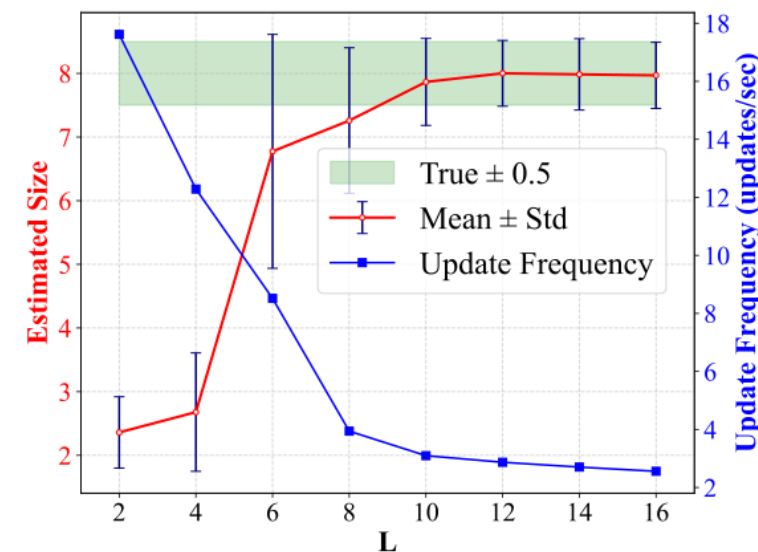
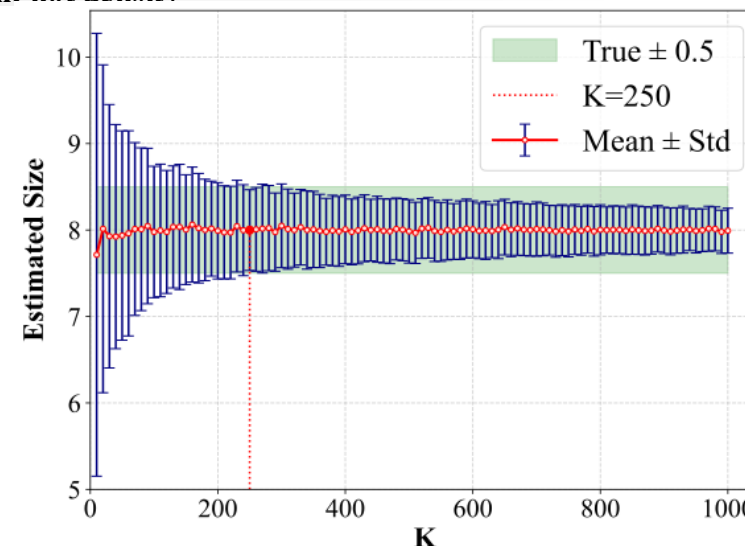


Accuracy
Overhead

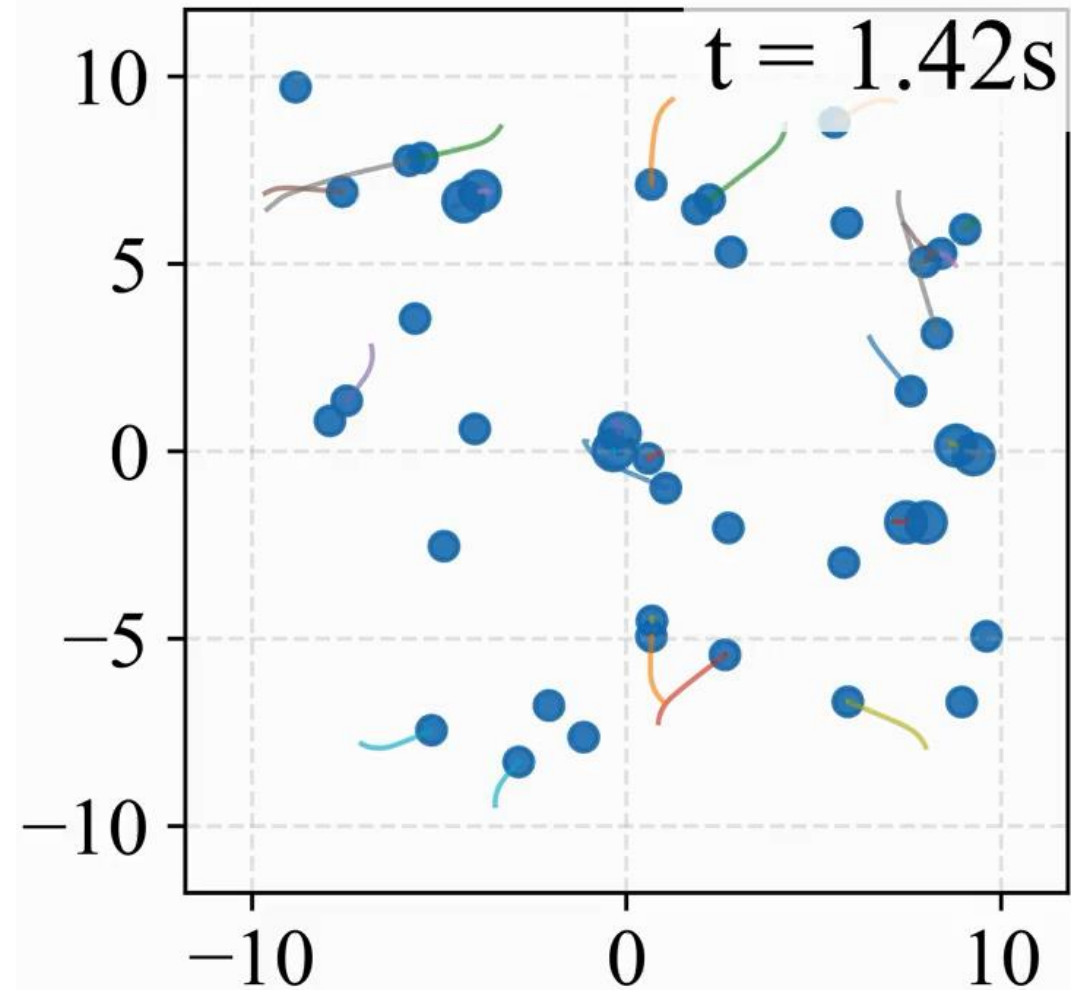
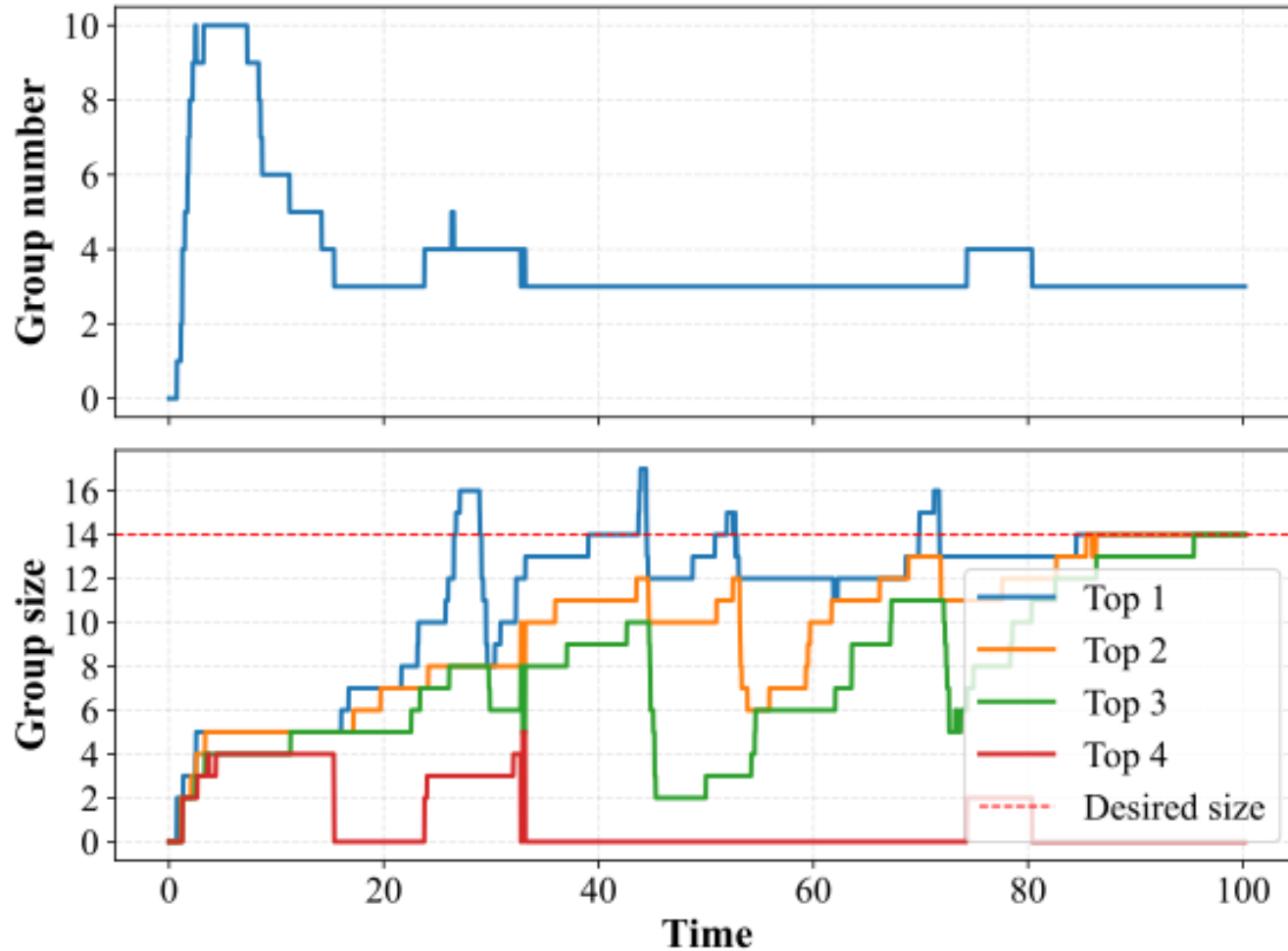
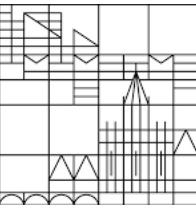
Group size

Round_id

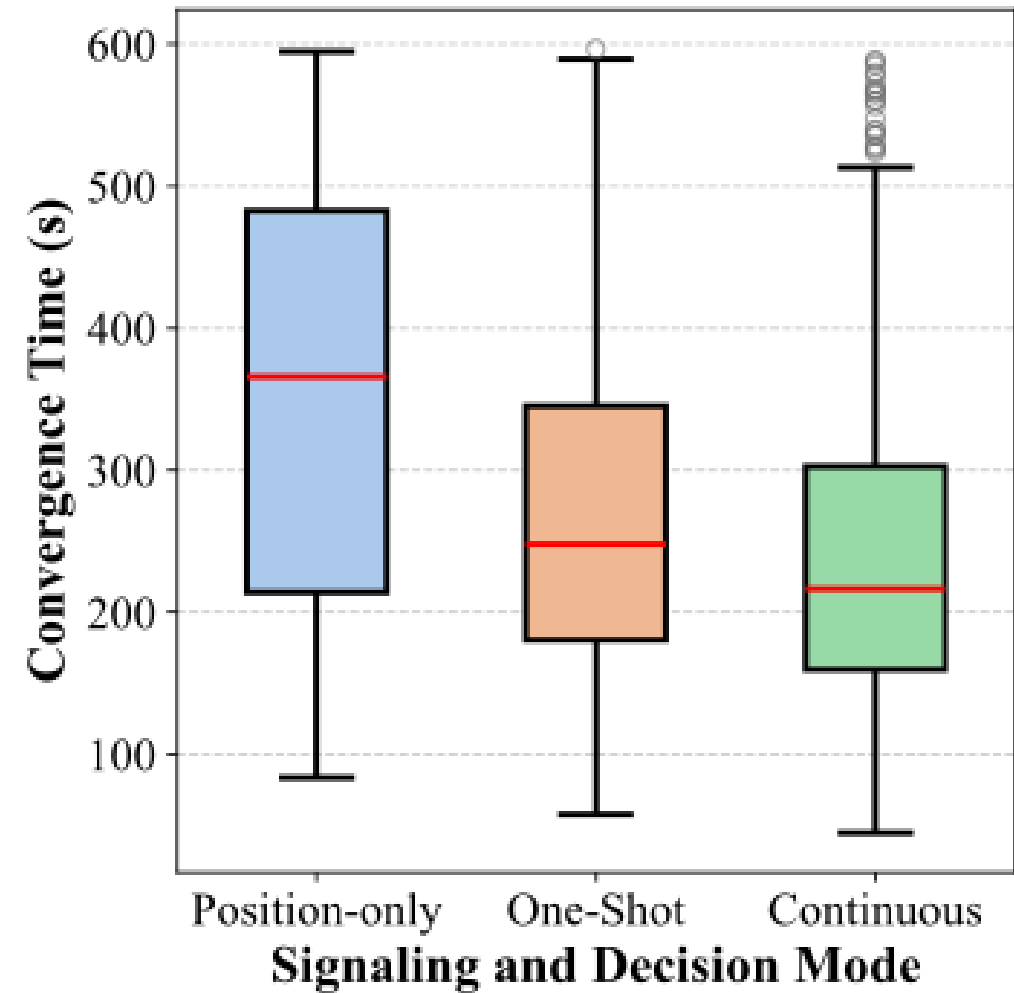
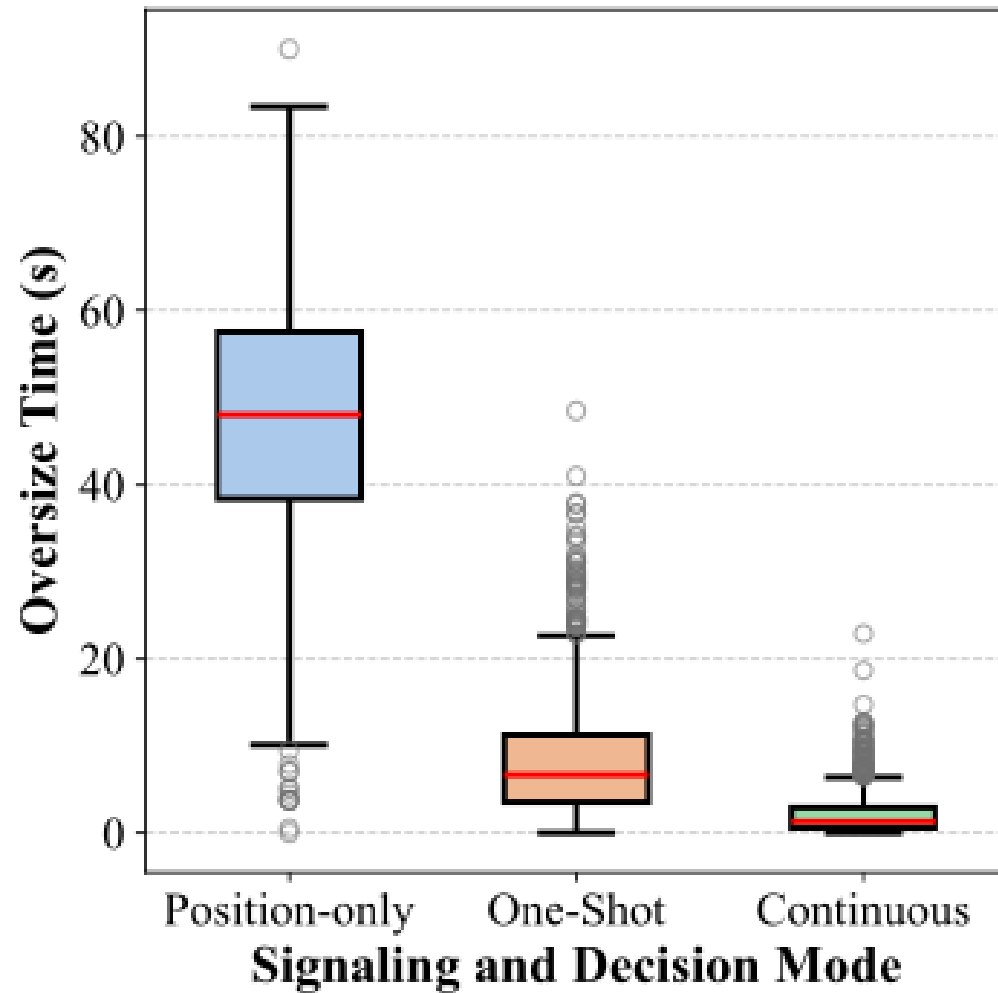
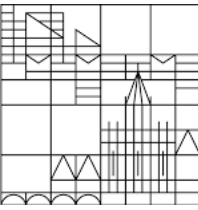
Accuracy
Frequency



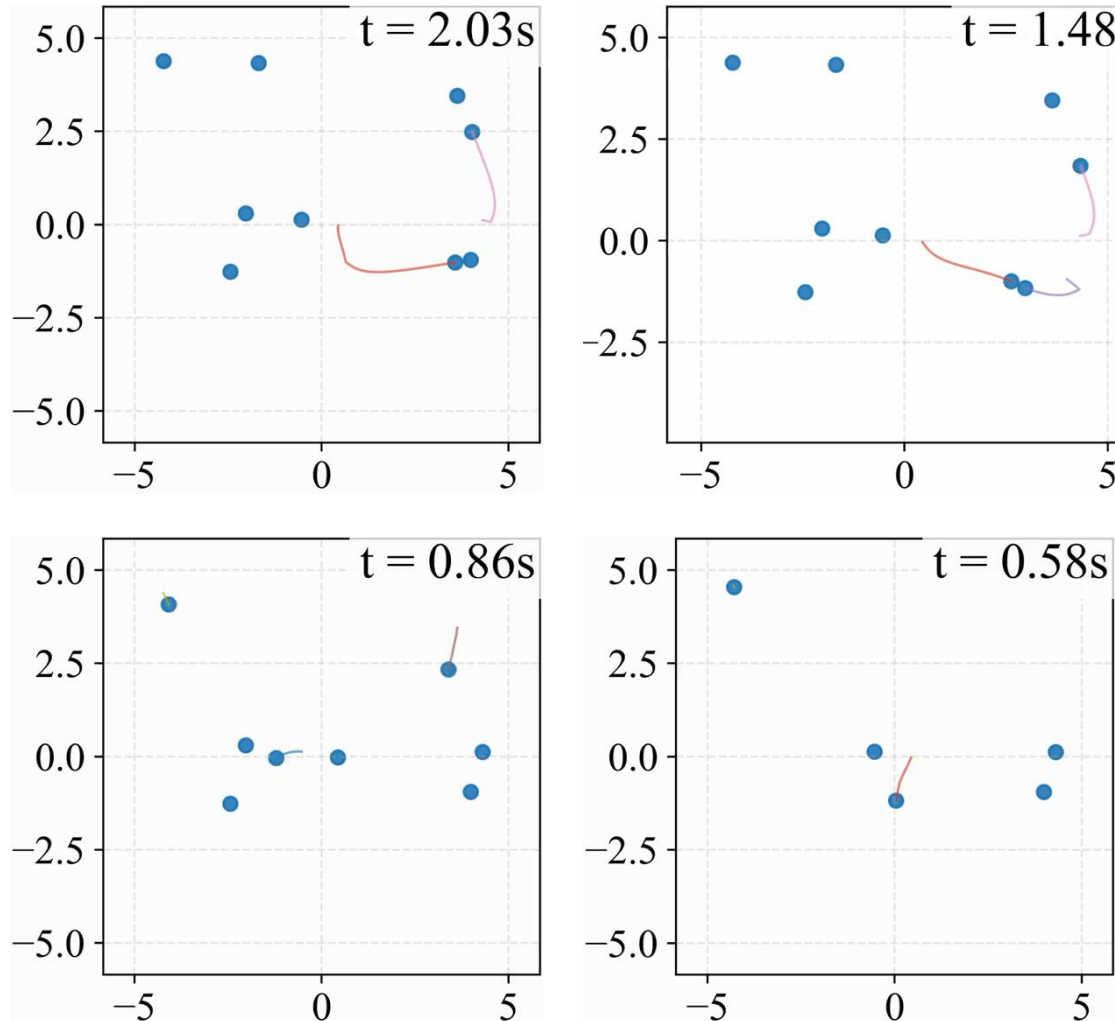
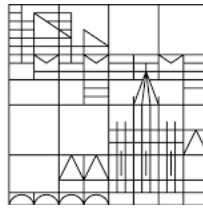
Stability and Convergence



Oversize and Convergence Time

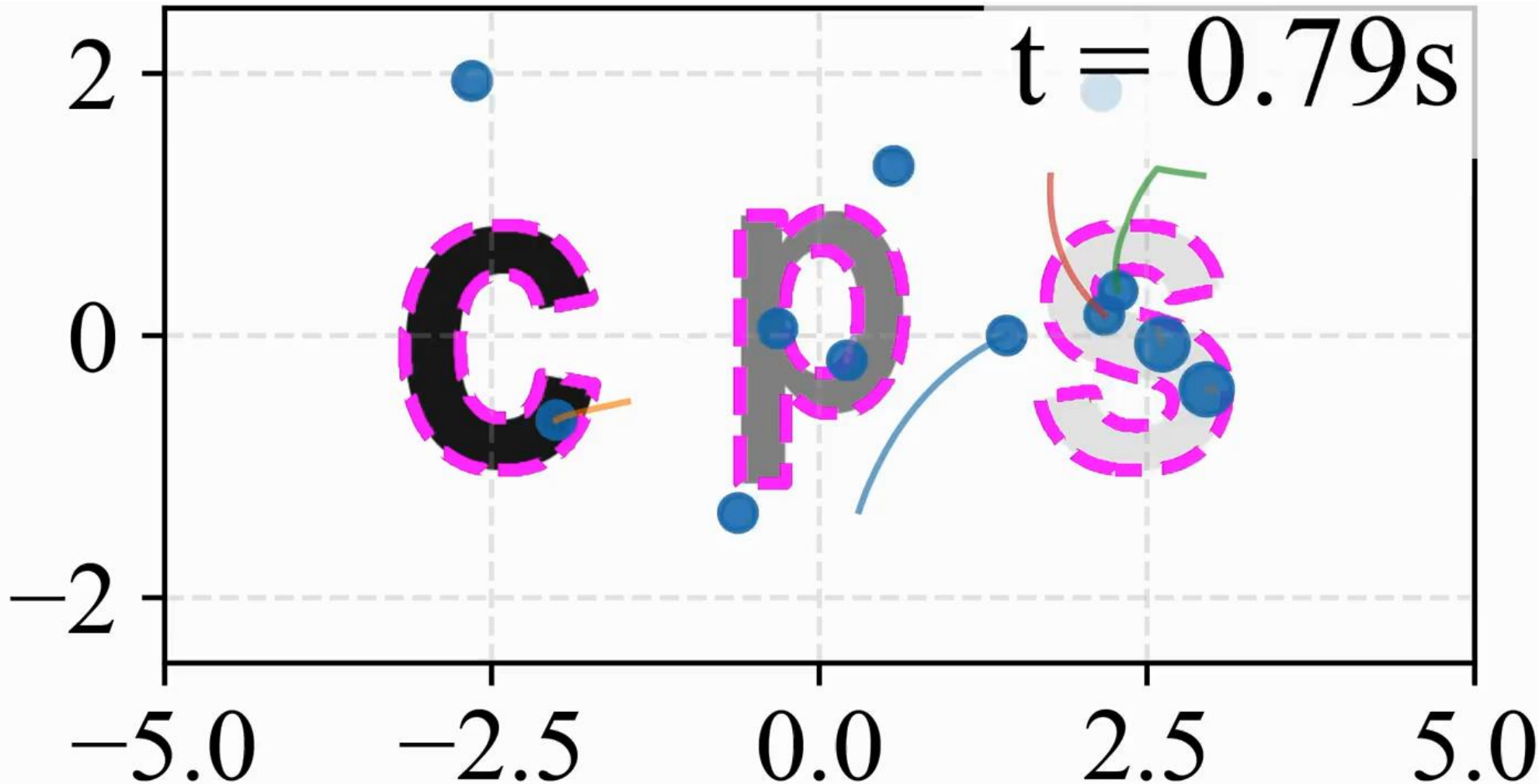
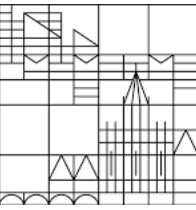


Benchmark



Metric	Value
Number of agents	9
Target subgroup sizes	2, 3, 4
Arena size	10 × 10 m
Positional signaling range	5 m
Gossip communication range	1 m
Maximum speed	1 m/s
Experimental duration (average)	226.03 s
Experimental duration (minimum)	180.34 s
Experimental duration (maximum)	270.52 s
Total number of trials	100
Success rate	87%
Max completion time (successful trials)	237.01 s
Mean completion time	42.14 s
Standard deviation	37.12 s

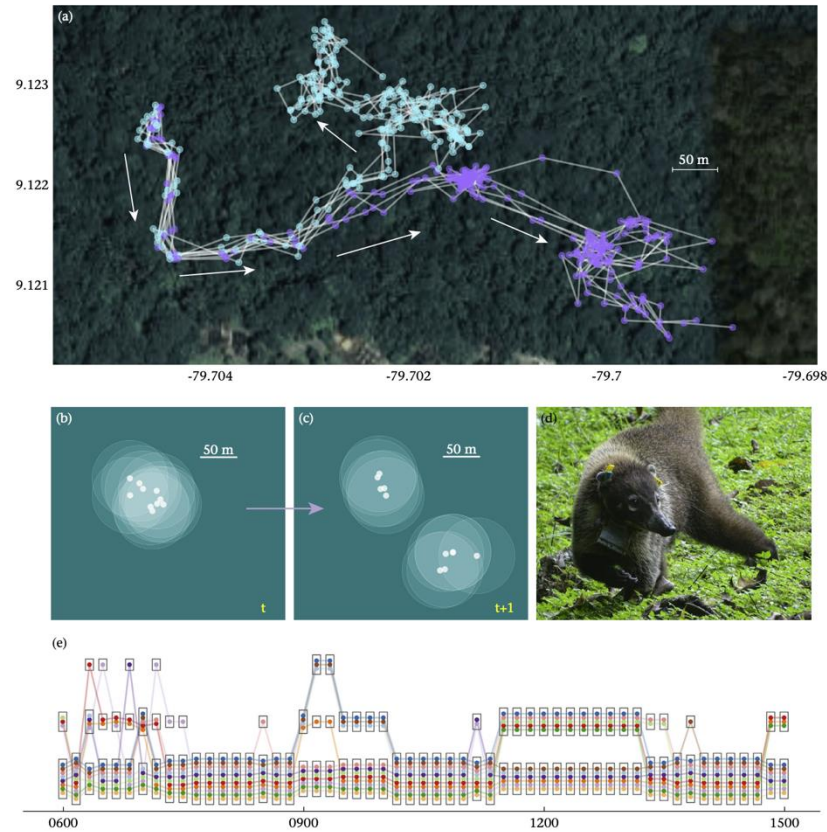
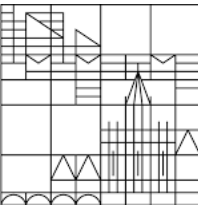
Spatially Determined Tasks



• **C:5**

• **P:7**

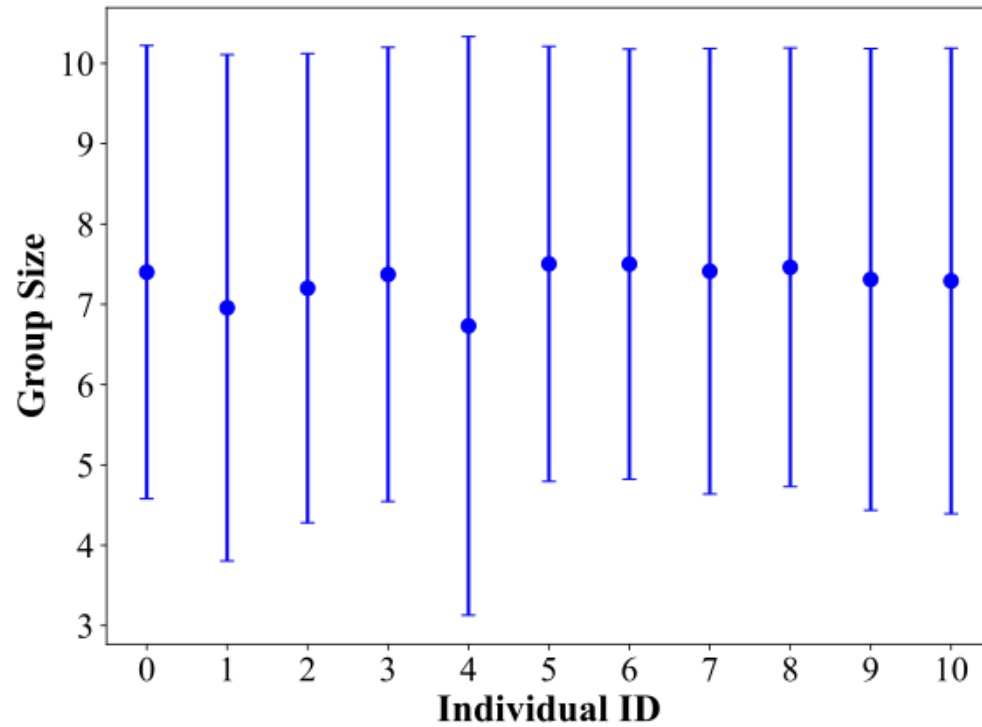
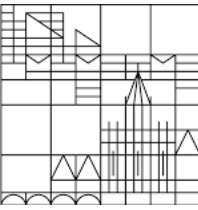
• **S:9**



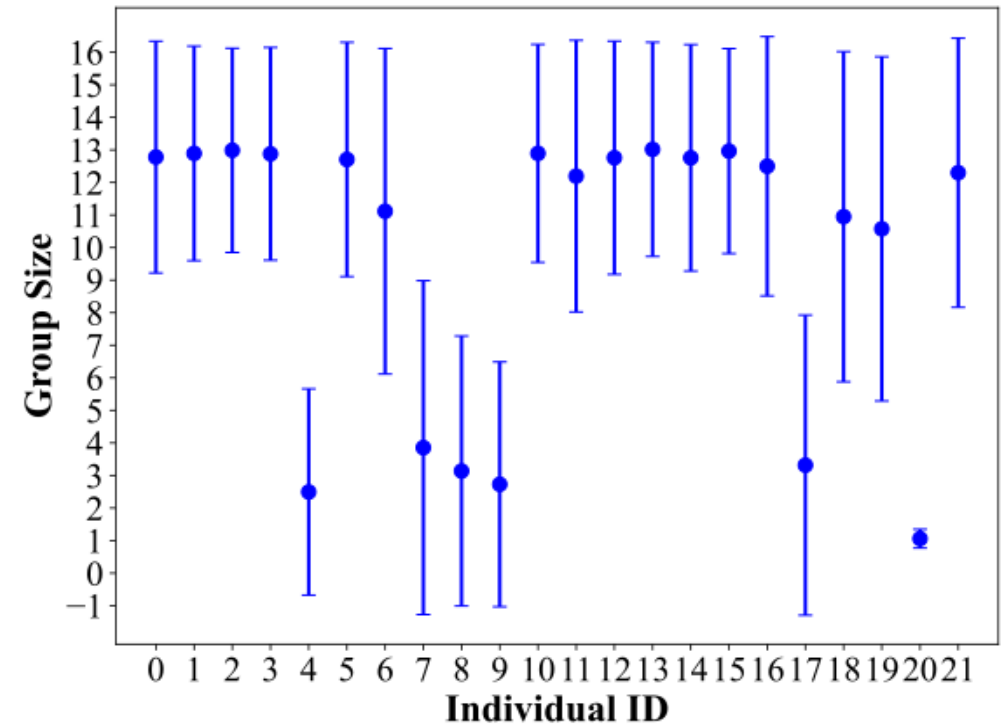
Whole group tracking and quantifying fission-fusion dynamics^[1]

[1] Grout E M, et al. Whole group tracking reveals that relatedness drives consistent subgrouping patterns in white-nosed coatis[J]. *Animal Behaviour*, 2024.

Set up

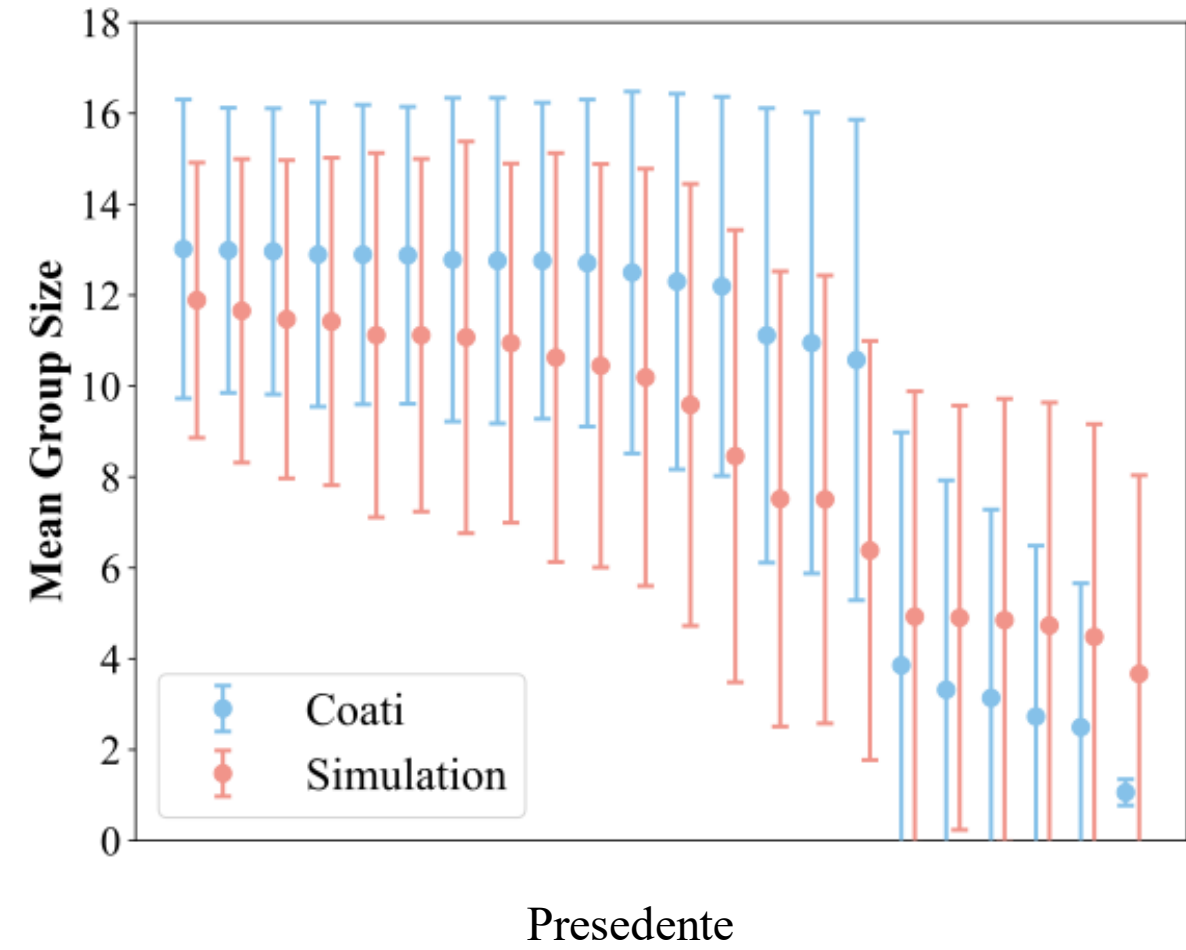
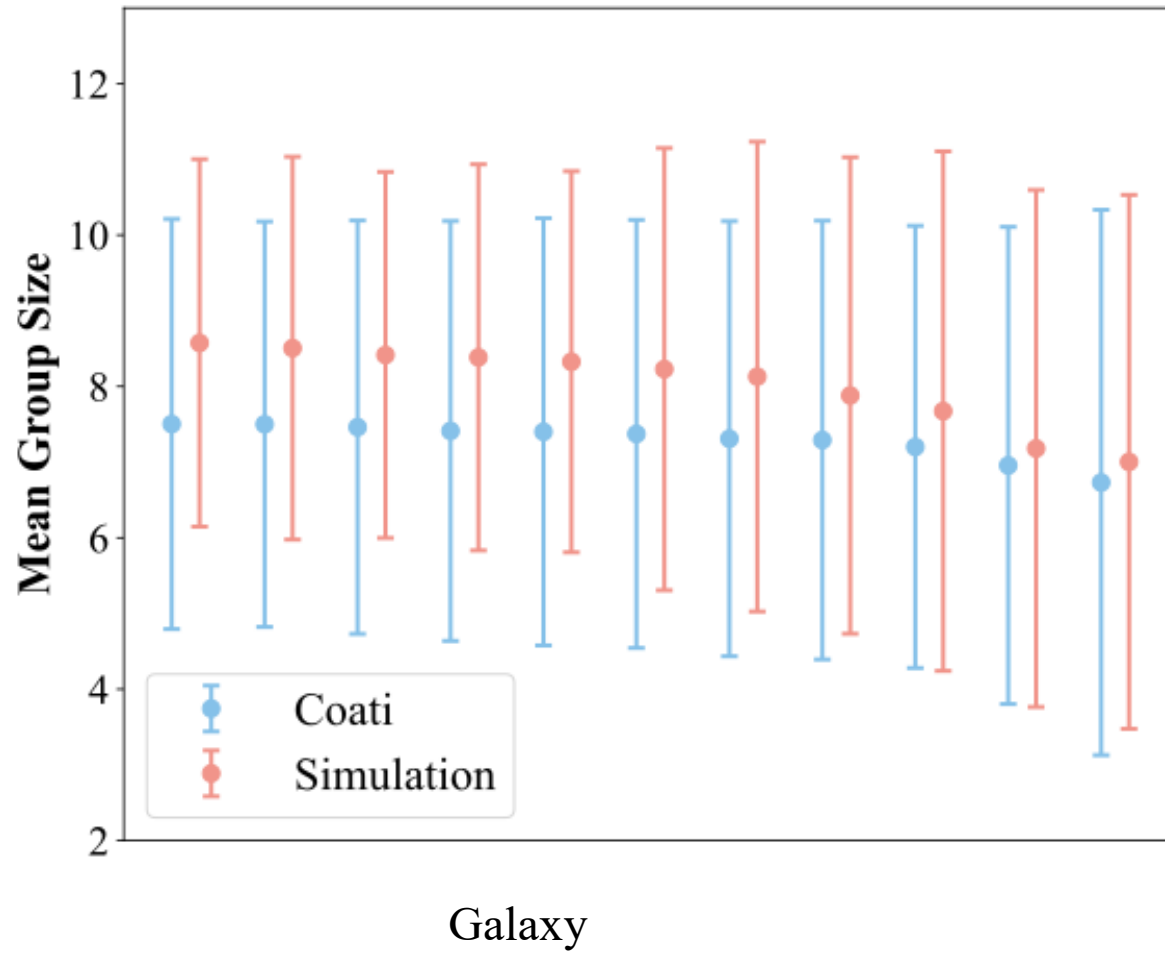
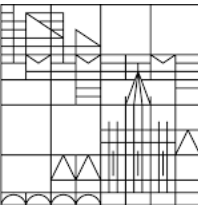


Galaxy

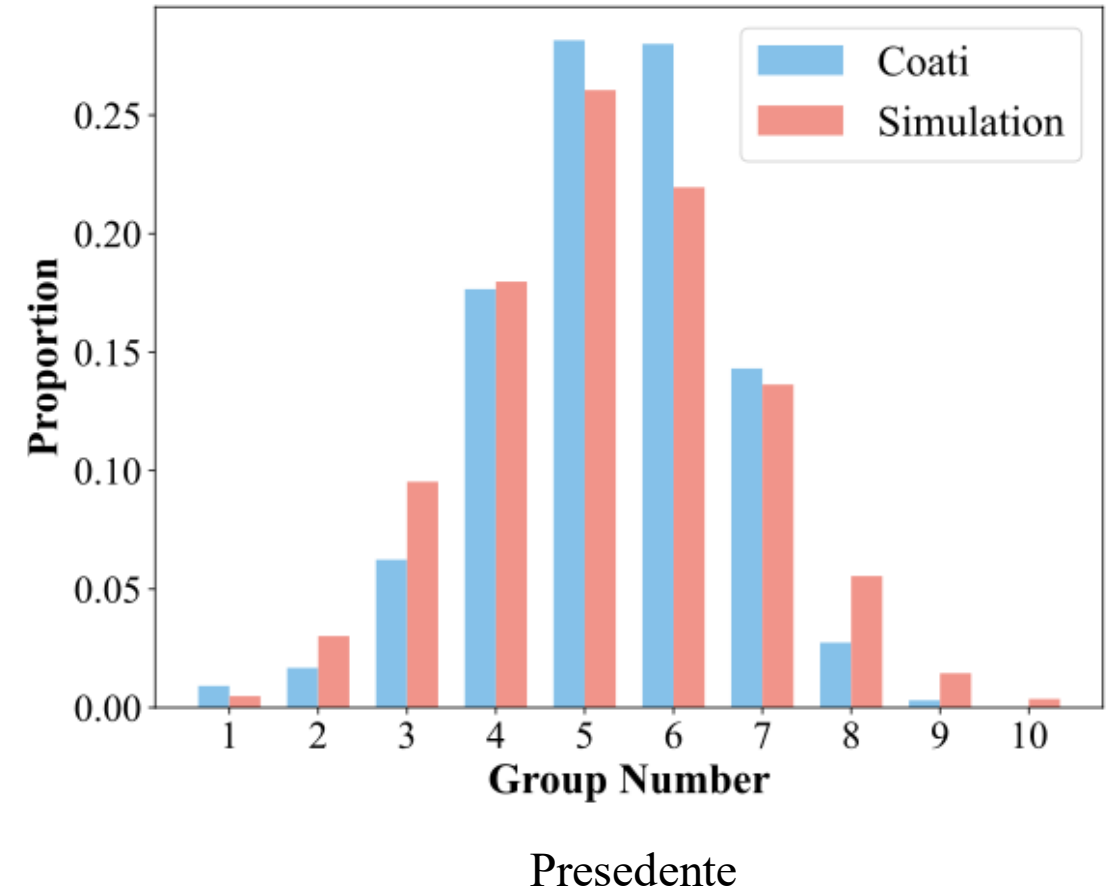
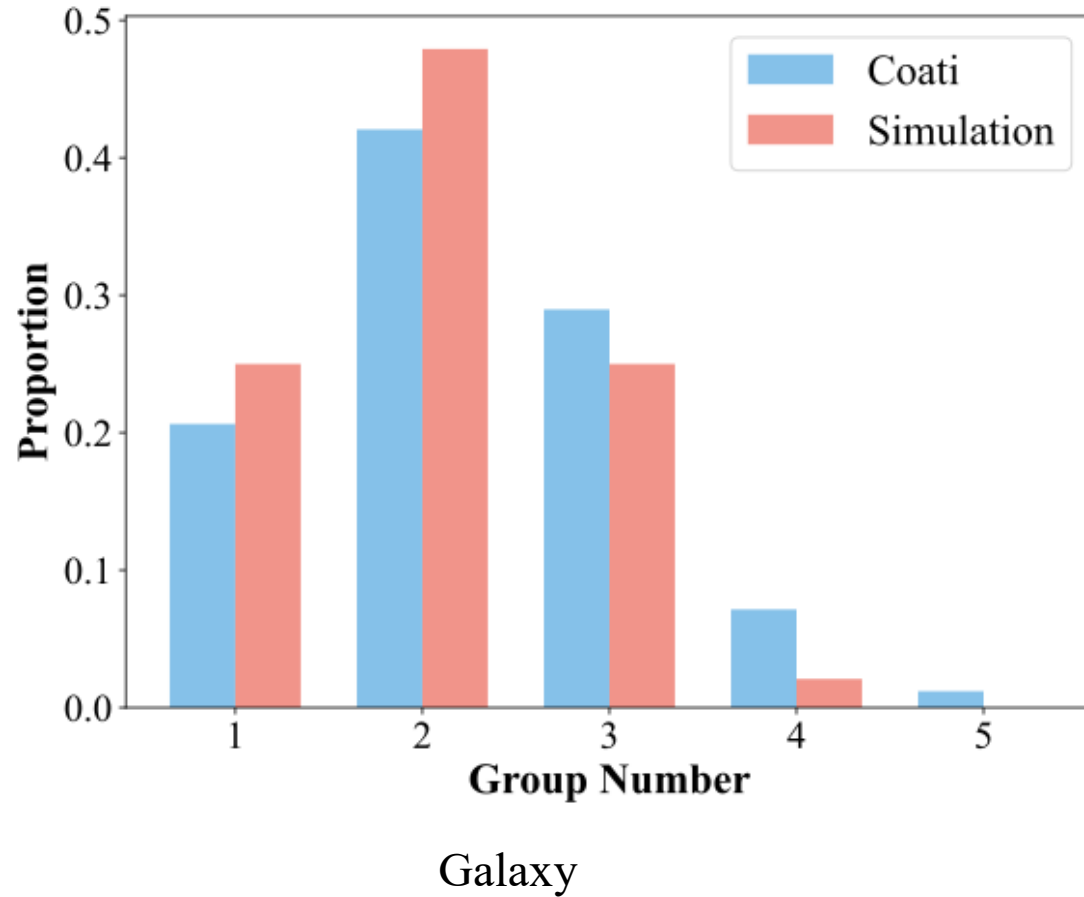
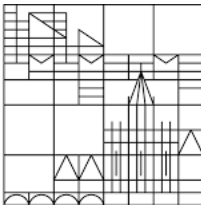


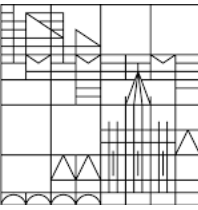
Presedente

Group size

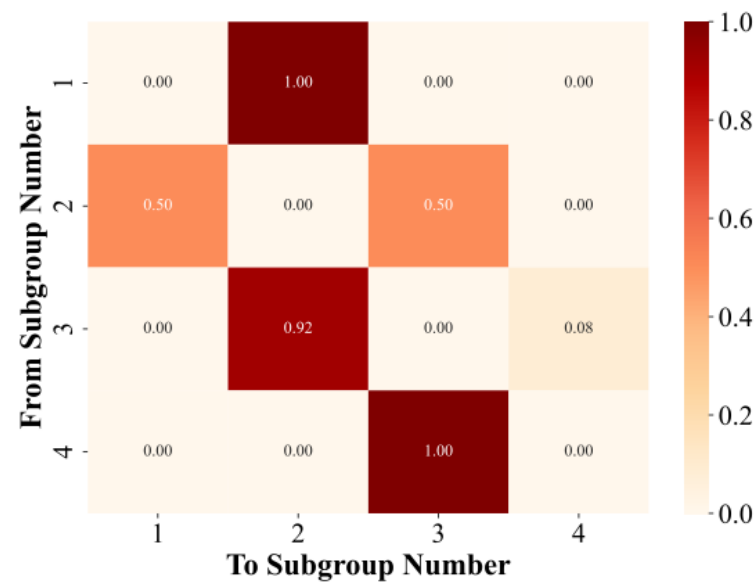


Group number



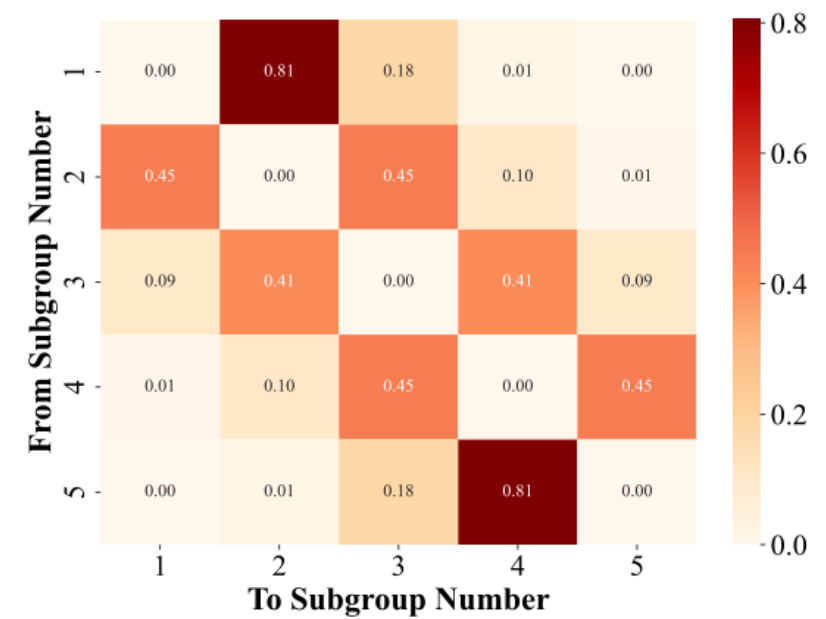


Coati



Simulation

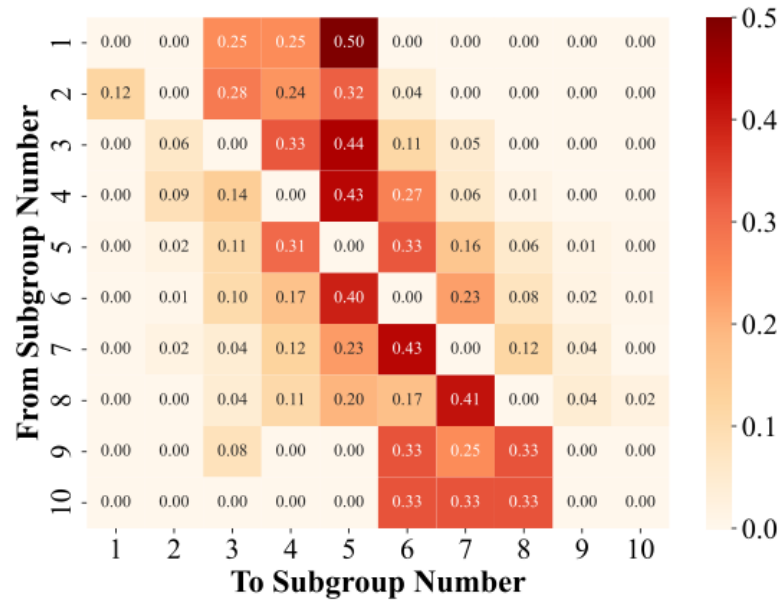
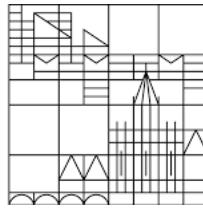
JS : 0.0324



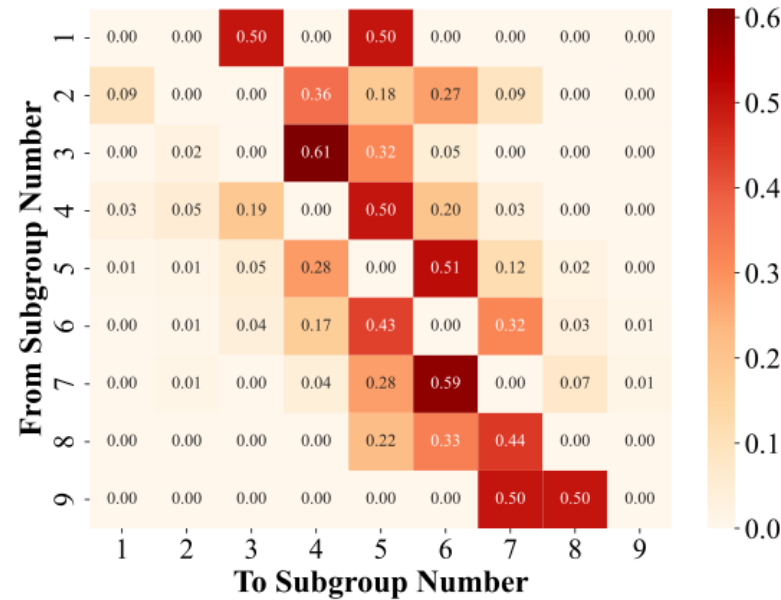
Gaussian

JS : 0.2309

Presedente

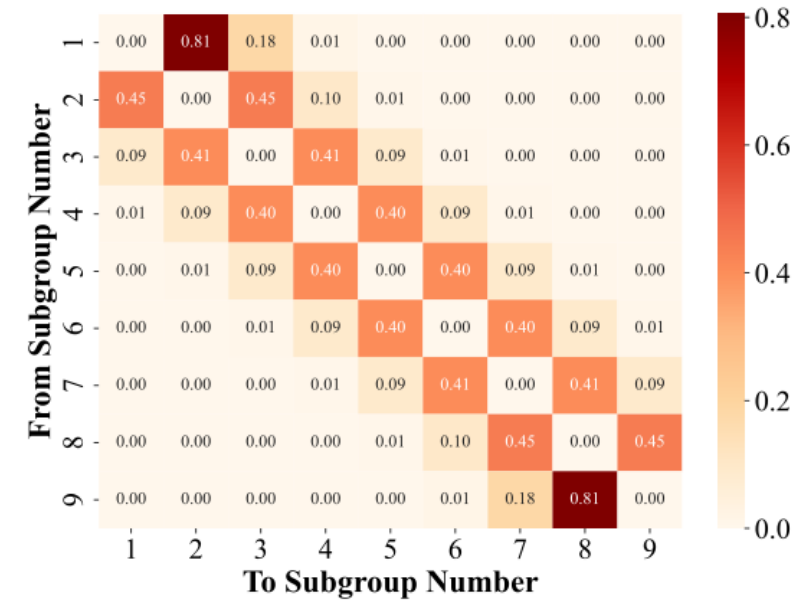


Coati



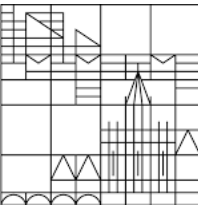
Simulation

JS :0.1069



Gaussian

JS : 0.2309



Danke